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Effect of molten pool thermal-flow behavior on microstructure uniformity during laser welding for unequal-gap Ti-6Al-4V alloy structure

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Abstract

Unequal-gap conditions in the laser welding of thin-walled Ti-6Al-4V alloy structures are inevitable due to uneven heat distribution and thermal distortion. Adaptive laser welding, which dynamically modulates welding parameters in response to changing conditions, provides an effective solution for fabricating structures with unequal-gaps. However, variations in gap size modify molten pool dynamics, leading to uneven surface formation and reducing the microstructural uniformity of the welded joint. In this study, a gradient-based parameterization strategy was implemented to enable adaptive laser welding of Ti-6Al-4V structures with variable gaps. The thermal-flow coupled model for unequal-gap

Outline

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welding was also established, wherein the separation between solid and gas phases increases progressively along the welding direction. The influence of molten pool flow behavior on microstructural uniformity under varying gap conditions was analyzed. A critical challenge in the aerospace sector is maintaining joint integrity when welding thin structures with unequal-gaps. As the gap increases, the overall molten pool shape changes from an X-shape to a Y-shape, while Marangoni convection at the bottom gradually disappears. Consequently, the maximum fluid flow velocity decreases by 40.3% (from 1.092m/s to 0.652m/s). The reduced flow rate enhances the cooling rate of the molten pool, leading to staggered growth of α and α' phases in localized regions. Compared with the zero-gap condition, the grain size of the titanium alloy decreases by 9.6% at a gap of 0.2mm.

Keywords

Ti-6Al-4V; Laser welding; Unequal-gapstructure; Microstructure uniformity; Molten pool flow

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1. Introduction

The advancement of the aerospace industry is driven by the continuous evolution of structural materials and the parallel development of fabrication technologies [1], [2]. As modern aircraft design increasingly prioritizes lightweight structures and geometric complexity, precision manufacturing of high-performance components has become a critical enabler of aerospace engineering [3]. Titanium alloys are extensively utilized in aerospace, automotive, marine, and biomedical applications owing to their high specific strength, favorable modulus-to-weight ratio, excellent corrosion resistance, and good weldability [4], [5], [6], [7], [8]. Among them, Ti-6Al-4V, a dual-phase $\alpha+\beta$ alloy, is the

CRediT authorship contribution statement

Declaration of competing interest

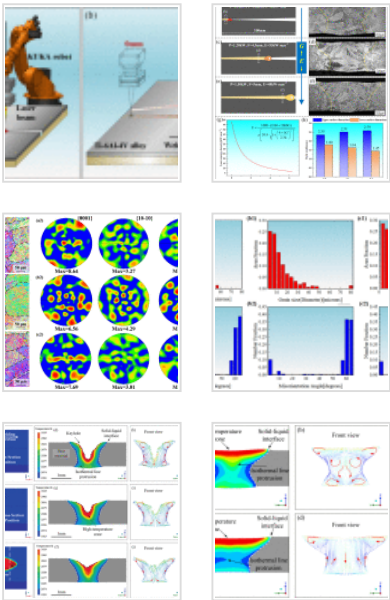
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most widely employed titanium alloy in the aerospace sector [9]. The use of titanium alloys dates back to the Cold War, most prominently in the SR-71 Blackbird reconnaissance aircraft, which employed large quantities of titanium to sustain flight at speeds above Mach 3.35 [10]. In contemporary aerospace vehicles, titanium alloy components serve as primary load-bearing and critical structural elements. Owing to their complex geometries and stringent precision requirements, these components present substantial challenges for conventional fabrication methods in achieving integrated, high-efficiency production [11], [12]. Welding technology enables rapid joining of high-precision components [13]. Developing advanced welding methods for efficient, high-quality joining of complex titanium alloy structures remains a central technological challenge in aerospace manufacturing.

Laser welding has attracted considerable attention for fabricating critical thin-walled structures, such as aircraft skin panels, owing to its high energy density and rapid processing speed [14], [15], [16]. However, limited machining precision and post-weld distortion complicate assembly. Excessive joint gaps often necessitate segmenting the weld into multiple passes, which increases weld seams, reduces efficiency, and compromises joint mechanical integrity [17], [18], [19]. These limitations constrain the wider application of laser welding for geometrically complex titanium alloy components.

To address the challenges of welding geometrically complex titanium alloy components, extensive research has focused on optimizing process parameters to enhance joint quality and reliability. Cheng et al. applied high-speed imaging to examine spatter behavior during the laser welding of 4mm-thick titanium alloys, demonstrating a clear correlation between process parameters and the dynamic behavior of metal vapor in the keyhole region [20]. Similarly, Lu et al. improved weld quality by optimizing the laser beam oscillation pattern, which reduced spatter formation during welding [21]. Yu et al. investigated the mechanism of laser oscillating patterns on keyhole stability and droplet dynamics in laser-MIG hybrid welding, demonstrating the effective suppression of porosity and surface defects in titanium alloy welds [10]. Li et al. investigated the laser welding of titanium alloys under different inclination angles, clarifying the distortion

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mechanisms that influence the trajectory of β -phase columnar grain growth under large-angle conditions [22]. Jallouf et al. evaluated surface defects in laser-welded titanium alloy joints, quantifying undercut features and predicting failure probability using statistical models [23]. Bikash et al. analyzed residual stress and deformation in thin titanium alloy weldments, emphasizing the influence of phase transformation on stress accumulation and post-weld distortion [24]. Duan et al. investigated filler metal transfer mechanisms and molten pool dynamics in laser wire-filling welding, demonstrating the production of high-quality welds in 2mm-thick titanium alloy structures with variable joint gaps [25]. Yu et al. investigated arc behavior and molten pool flow in narrow-gap oscillating laser-arc hybrid welding, demonstrating the fabrication of stable, low-porosity titanium alloy joints with enhanced microstructural uniformity [26].

The optimization of process parameters plays a pivotal role in enhancing weld quality during laser welding of titanium alloys, directly influencing molten pool behavior, microstructural evolution, and the mechanical performance of welded joints. However, they also highlight the inherent limitations of experimental methods in fully elucidating the complex physical mechanisms governing molten pool dynamics and microstructural evolution. To better understand the thermal-flow behavior in welding complex geometries, numerical simulation has become an indispensable tool for characterizing molten pool flow, heat transfer, and solidification. Yan et al. employed a dual-laser-beam bilateral synchronous welding method to join titanium alloy structures, and through integrated simulations, revealed the formation mechanisms of coarse columnar grains in the weld metal [27]. Wang et al. performed numerical simulations of molten pool flow and thermal gradient distribution during ring-core adjustable laser welding of narrow-gap titanium alloy plates [28]. The results showed that increasing the groove gap transformed the weld cross-sectional morphology from narrow and deep to wide and shallow. Similarly, Wu et al. used numerical simulation to predict the stress-strain response during laser welding, with outputs closely matching experimental data and accurately forecasting fracture strain, thereby demonstrating the capability of computational models to predict post-weld mechanical behavior [29].

In the laser welding of thin-walled Ti-6Al-4V structures, non-uniform joint gaps inevitably arise from uneven heat distribution and thermal distortion. Furthermore, this process inevitably induces significant interfacial stress, which remains a critical issue affecting the structural integrity of the joints [30], [31]. However, a comprehensive understanding of how these dynamically varying gaps influence molten pool thermal-flow dynamics and subsequent microstructural evolution remains elusive. Most existing numerical models predominantly assume a constant gap geometry, neglecting the transient thermal behaviors induced by gap fluctuations. To address this, this study systematically investigates the processes and numerical simulations associated with unequal-gap conditions. A detailed analysis is conducted to elucidate the coupled molten pool flow dynamics and microstructural evolution, which serves as an essential prerequisite for future evaluations of interfacial stress. These findings provide new insights into adaptive laser welding and offer a robust framework for achieving consistent, high-quality welds in thin-walled structures with variable joint geometries.

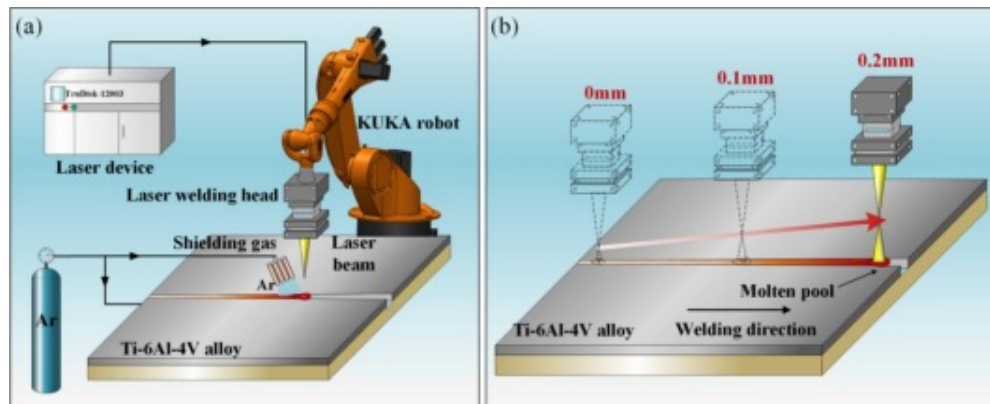
2. Experimental details

The Ti-6Al-4V alloy plate (100mm×50mm×1.2mm) in a forged condition was used as the substrate. The chemical composition of the alloy is provided in Table 1. To minimize contamination and improve weld quality, surface preparation was performed within 2h before welding. The procedure included mechanical removal of the native oxide layer using abrasive paper, degreasing with acetone, and subsequent air drying to obtain a clean, moisture-free surface.

Table 1. Chemical composition of Ti-6Al-4V Alloy (wt.%).

Chemical Composition	Fe	C	H	O	N	Al	V	Ti
wt.%	≤0.30	≤0.10	≤0.015	≤0.20	≤0.05	5.5–6.8	3.5–4.5	Bal.

Laser welding experiments were performed using a Trudisk-12003 fiber laser system (Trumpf, Germany) integrated with a KR60HA high-precision robotic welding arm to maintain stable and repeatable weld trajectories. Ti-6Al-4V alloy plates (1.2mm thick) were rigidly secured in a custom-designed clamping fixture to minimize distortion. The initial joint gap was set to 0mm and gradually increased to 0.2mm along the weld path to simulate unequal-gap conditions. A schematic of the adaptive laser welding setup for Ti-6Al-4V alloy under unequal-gap configurations is shown in Fig. 1.



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Fig. 1. (a) Schematic of variable-gap laser welding equipment; (b) schematic illustration of laser welding under unequal-gap conditions.

To simulate unequal-gap conditions, a total welding length of 100mm was established, with the joint gap linearly increasing from 0 to 0.2mm. To strictly maintain this precise gap profile and mitigate thermal distortion during the welding process, the Ti-6Al-4V plates were rigidly constrained using a customized heavy-duty clamping fixture equipped with precision spacer blocks. Throughout the process, the shielding gas flow rates were maintained constant, utilizing a positive flow of $20\text{L}\cdot\text{min}^{-1}$ and a negative flow of $5\text{L}\cdot\text{min}^{-1}$. To implement the adaptive welding strategy, a high-precision laser vision system was employed to dynamically capture the initial weld seam position and the

corresponding joint gap. This vision system possesses a recognition accuracy of ± 0.02 mm, ensuring highly reliable gap detection. The acquired geometric data were instantaneously transmitted to the welding control system to determine the start and end positions and generate the optimal real-time process parameters (summarized in Table 2). Notably, the overall response latency of this parameter feedback control system is strictly maintained within 50 ms, effectively satisfying the stringent requirements for practical industrial applications. Guided by this closed-loop adaptive system, the laser power and defocusing distance were linearly increased in tandem with the widening joint gap. Consequently, the laser spot radius expanded dynamically to accurately match the instantaneous gap size. This real-time modulation of energy input and spot geometry ensured a stable keyhole formation, ultimately guaranteeing consistent macroscopic formation and weld quality along the entire unequal-gap joint.

Table 2. Experimental parameters for laser adaptive welding.

Weld joint gap/mm	Laser power/kW	Defocus/mm	Welding speed/m·min ⁻¹
0~0.2	1.1~1.3	4~5	1.2

3. Modeling

During laser welding of Ti-6Al-4V alloy, the molten pool dynamics, including high-temperature liquid metal flow and keyhole fluctuations, are governed by interaction with the laser heat source. To capture these phenomena, a thermal-flow coupled model was developed that accounts for heat and mass transfer under the influence of gravity, enabling accurate numerical simulation of the welding process. The essential thermophysical properties of the base metal, which serve as critical input parameters for the model, are summarized in Table 3 and were obtained from established literature sources [32], [33].

Table 3. Thermophysical properties of Ti-6Al-4V alloy [37].

Physical	Unit	Value
Liquidus temperature	K	1878
Solidus temperature	K	1928
Boiling point	K	3133
Liquidus density	$\text{kg}\cdot\text{m}^{-3}$	4000
Solidus density	$\text{kg}\cdot\text{m}^{-3}$	4510
Thermal conductivity of liquid	$\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$	29
Thermal conductivity of solid	$\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$	21
Specific heat of liquid	$\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$	730
Specific heat of solid	$\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$	670
Dynamic viscosity	$\text{kg}\cdot\text{m}^{-1}\cdot\text{s}^{-1}$	0.005
Coefficient of thermal expansion		8×10^{-6}
Thermal emissivity		0.4
Ambient temperature	K	300
Convective heat transfer coefficient	$\text{W}\cdot\text{K}^{-1}\cdot\text{m}^{-2}$	15
Latent heat of fusion	$\text{J}\cdot\text{kg}^{-1}$	1.12×10^6
Latent heat of evaporation	$\text{J}\cdot\text{kg}^{-1}$	1.49×10^6
Surface tension	$\text{N}\cdot\text{m}^{-1}$	1.4
Temperature gradient of surface tension	$\text{N}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$	-0.26×10^{-3}

To optimize computational efficiency while preserving macroscopic physical fidelity, the complex multi-physics of the laser welding process, with encompassing heat transfer,

fluid flow, and solid–liquid phase transitions, are streamlined using a set of rigorously grounded assumptions. The molten metal is modeled as an incompressible Newtonian fluid operating within a laminar flow regime, with the transient thermal domain initialized at an ambient temperature of 300K. To balance computational cost with predictive accuracy, density and dynamic viscosity are assumed constant. Conversely, specific heat and thermal conductivity are strictly treated as temperature-dependent to accurately resolve the thermal-flow behavior, which is predominantly driven by temperature gradients and the Marangoni effect. Regarding boundary interactions, the model incorporates the dominant effect of recoil pressure on the molten pool surface. It is important to delineate that while the aerodynamic scattering, absorption, and shielding gas convection induced by the metal evaporation plume are neglected to simplify the numerical framework, the boundary condition for the latent heat of evaporation (q_{evap}) is strictly retained. This deliberate distinction ensures the accurate capture of energy dissipation associated with the transient keyhole dynamics.

Furthermore, to guarantee numerical robustness and reliability, a rigorous mesh independence test was conducted prior to the full-scale simulations. Comparative analyses among coarse (0.1 mm), intermediate (0.05 mm), and fine (0.025 mm) grids revealed that the selected minimum grid spacing of 0.05 mm yielded a maximum temperature deviation of less than 2% relative to the finest mesh, while simultaneously reducing computational time by approximately 60%. This focused computational framework ensures that the governing equations remain highly tractable without sacrificing the spatial accuracy of the underlying thermal-flow phenomena.

3.1. Governing equations

To accurately simulate the laser welding of the Ti-6Al-4V alloy, the internal thermal-flow dynamics with including metal transport and thermal energy dissipation are modeled based on the three-dimensional conservation laws. These governing mechanisms, which establish the theoretical basis for our numerical predictions, are defined as follows:

Mass conservation equation [34]:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (1)$$

In this expression, t denotes the welding time, and the velocity field $\mathbf{u}$ is decomposed into components k , l , and n corresponding to the three-dimensional spatial axes.

Momentum conservation equation [35]:

X Direction:

$$\rho \left[\frac{\partial k}{\partial t} + (\mathbf{u}_{rel} \cdot \nabla) k \right] = -\frac{\partial P}{\partial x} + \mu \nabla^2 k + S_k \quad (2)$$

Y Direction:

$$\rho \left[\frac{\partial l}{\partial t} + (\mathbf{u}_{rel} \cdot \nabla) l \right] = -\frac{\partial P}{\partial y} + \mu \nabla^2 l + S_l \quad (3)$$

Z Direction:

$$\rho \left[\frac{\partial n}{\partial t} + (\mathbf{u}_{rel} \cdot \nabla) n \right] = -\frac{\partial P}{\partial z} + \mu \nabla^2 n + S_n \quad (4)$$

In these expressions, P represents the hydrodynamic pressure, while μ denotes the temperature-dependent dynamic viscosity. The term $\mathbf{u}_{rel}$ is defined as the relative velocity vector of the molten metal with respect to the translating laser heat source, ensuring the coupling of the welding speed into the convective transport terms. The terms S_k , S_l , and S_n designate the momentum source components, which incorporate the combined effects of buoyancy, Marangoni flow, and recoil pressure acting along the respective Cartesian axes.

Energy conservation equation [36]:

$$\rho \left[\frac{\partial H}{\partial t} + (\mathbf{u}_{rel} \cdot \nabla) H \right] = \nabla \cdot (k \nabla T) + S_E \quad (5)$$

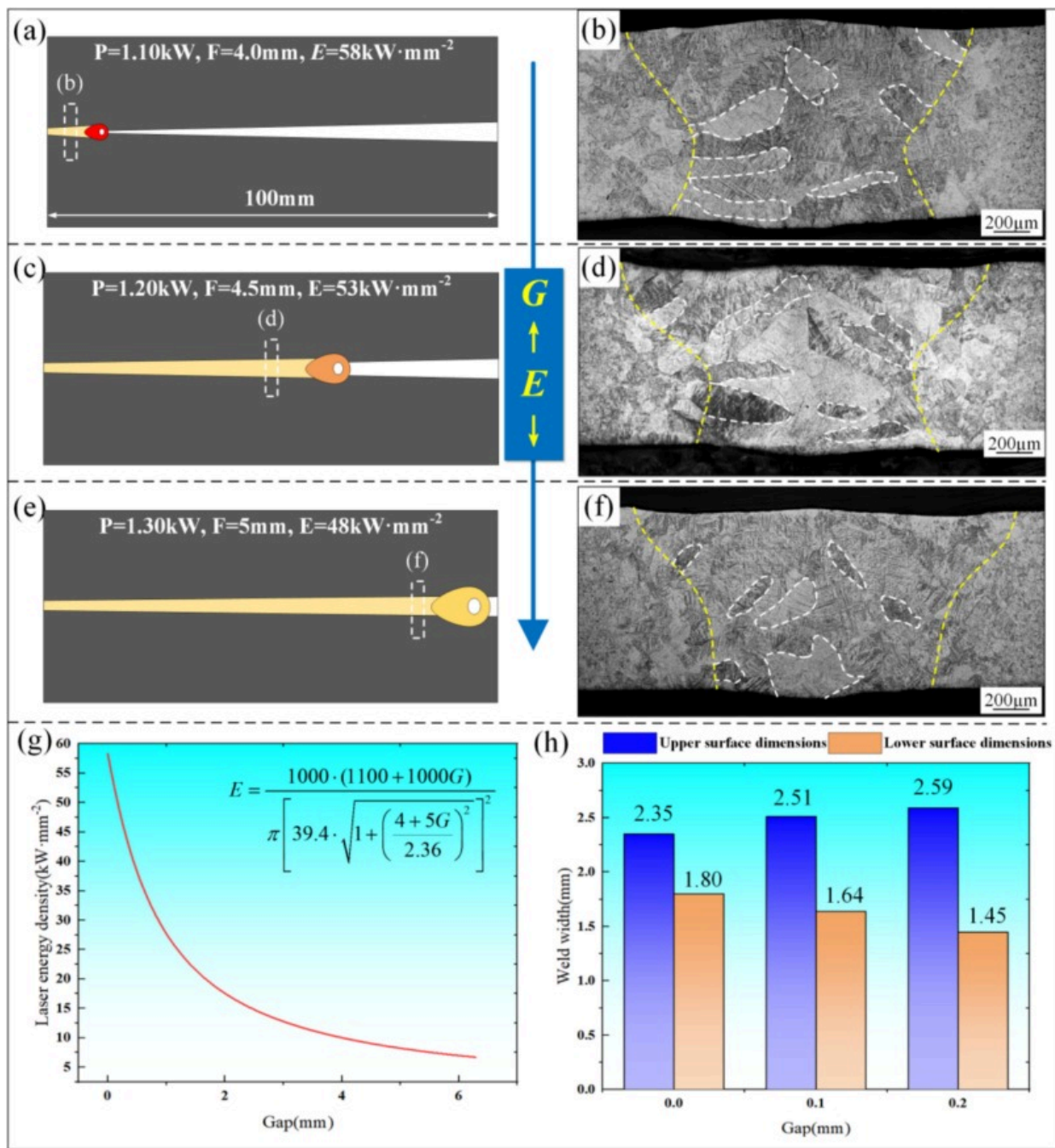
In this thermodynamic system, k represents the material's thermal conductivity, which varies with temperature, and H designates the total mixture enthalpy of the system. The comprehensive energy source term, S_E , accounts for the heat input and various thermal losses, formulated as [39]:

$$S_E = Q - m_v L_v - \rho \frac{\partial(\Delta H)}{\partial t} - \rho(u_{rel} \cdot \nabla) \Delta H \quad (6)$$

here Q signifies the volumetric laser heat input. To maintain physical fidelity, the model incorporates the thermal energy consumed during metal vaporization ($m_v L_v$) and the latent heat associated with phase transformations (ΔH), where m_v and L_v denote the mass evaporation rate and the latent heat of vaporization, respectively

3.2. Computational domains and boundary conditions

The geometric framework for the numerical simulation is depicted in Fig. 2. To optimize the computational resource allocation while maintaining high solution fidelity, a non-uniform meshing strategy was implemented. Specifically, the discretization was refined within the immediate vicinity of the laser-material interaction zone, where the magnitudes of thermal and stress gradients are most pronounced. In this optimized configuration, a minimum grid spacing of 0.05 mm was maintained at the weld centerline, leading to a total of 514,560 discrete elements to resolve the global thermo-mechanical response.



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Fig. 2. Morphology and distribution of β columnar grains in the fusion zone at different positions during adaptive laser welding with unequal-gap: (a, c, e) schematic of sampling position#1, position#2, position#3; (b, d, f) microstructure; (g) calculation method of laser power density; (h) weld seam dimensions.

During the laser welding procedure, the specimen surface undergoes a complex energy exchange governed by incident laser radiation, convective cooling, thermal radiation to the environment, and phase-change-induced evaporation. The integrated energy balance at the material interface is mathematically defined as follows [39]:

$$k \frac{\partial T}{\partial \vec{n}} = Q - q_{conv} - q_{rad} - q_{evap} \quad (7)$$

In this thermodynamic balance, $\vec{n}$ is the laser heat source, while the loss components-convective heat transfer (q_{conv}), radiative emission (q_{rad}), and evaporative cooling (q_{evap})-are formulated as:

$$q_{conv} = h_c(T - T_0) \quad (8)$$

$$q_{rad} = \varepsilon \sigma (T^4 - T_0^4) \quad (9)$$

$$q_{evap} = m_v L_v \quad (10)$$

In these expressions, T and T_0 designate the instantaneous surface temperature and the surrounding ambient temperature, respectively. The parameters h_c , ε , and σ denote the convective heat transfer coefficient, the surface emissivity, and the Stefan-Boltzmann constant, respectively.

3.3. Laser heat source model

To faithfully replicate the high-energy density and deep-penetration characteristics of the laser welding process, a hybrid heat source model was constructed. This configuration

integrates a Gaussian surface distribution with a Gaussian rotary volumetric source to account for the intense energy localization within the keyhole and the surrounding melt.

The surface energy distribution, which accounts for the initial laser-material interaction, is formulated as [40]:

$$q_s(x, y) = \frac{\eta_s Q_s}{\pi r_s^2} \exp\left(-\frac{C_s(x^2 + y^2)}{r_s^2}\right) \quad (11)$$

In this study, the effective radius r_s is dynamically coupled with the defocusing distance, η_s is the concentration coefficient of the Gaussian surface heat source. Specifically, the spatial evolution of r_s is dictated by the beam spot radius variations as defined in Eq. (16). The calibration parameter C_s was iteratively refined during the simulation to ensure the predicted weld morphology aligns with experimental cross-sections.

To resolve the energy deposition along the depth of the keyhole, a Gaussian rotary body heat source is utilized [41]:

$$q_r(x, y, z) = \frac{\eta_r Q_r}{\pi h r_e^2 (1 - e^{-3})} \exp\left(-\frac{C_r(x^2 + y^2)}{r_e^2 \ln(h/z)}\right) \quad (12)$$

where h denotes the effective penetration depth, Q_r is the surface heat Gaussian rotary body heat source, η_r is the concentration coefficient for the volumetric component, and C_r is the correction coefficient of the Gaussian rotary body heat source.

The total laser power Q absorbed by the Ti-6Al-4V substrate is partitioned between the surface and volumetric components, governed by the global absorption efficiency η :

$$\eta Q = \iiint (q_s + q_r) dV \quad (13)$$

While directly characterizing the ultra-fast real-time thermal-flow behavior in the molten pool remains experimentally challenging without customized high-speed synchrotron or in-situ quenching platforms, the excellent agreement between our numerical simulations and the final macroscopic weld morphology and microstructural distributions robustly validates the predictive capability of the proposed thermal-fluid model.

4. Result and discussion

4.1. Microstructural distribution characteristics

To compare microstructural differences in Ti-6Al-4V laser-welded joints under varying gap conditions, metallographic samples were prepared at three locations along the weld: the starting position, the center, and the end. The sampling locations and corresponding microstructures are shown in Fig. 2(a–f) and were designated as position #1, position #2, and position #3. The gaps, laser powers, and defocus distances differed at these positions. To unify the influence of these parameters on microstructure, the laser power density (E) was defined as the laser power per unit spot area. Because the laser spot area varies considerably with defocus distance, it was calculated as follows.

The spot diameter in the zero-defocus region (the beam waist diameter) depends on the collimating-lens and focusing-lens focal lengths and the fiber core diameter. It is calculated as follows:

$$\omega_0 = \frac{f}{f_c} \frac{\Phi}{2} \quad (14)$$

where ω_0 is the beam waist radius, f the focal length of the focusing lens, f_c the focal length of the collimating lens, and Φ the fiber core diameter.

Neglecting the energy distribution across the beam cross-section, the relationship between the beam spot radius and defocus can be expressed in terms of the Rayleigh length Z_R and the beam waist radius ω_0 .

$$Z_R = \frac{\pi \omega_0^2}{\lambda M^2} \quad (15)$$

$$\omega(F) = \omega_0 \sqrt{1 + \left(\frac{F}{Z_R}\right)^2} \quad (16)$$

where F denotes the defocus distance, $\omega(F)$ the beam spot radius at propagation distance F , λ the laser wavelength, and M^2 the beam quality factor.

Combining Eqs. (14), (15), (16) yields the final expression for the relationship between the laser beam spot radius and the defocus distance:

$$\omega(F) = \frac{\Phi}{2} \frac{f}{f_c} \sqrt{1 + \left(\frac{4F\lambda M^2 f_c^2}{\pi f^2 \Phi^2} \right)^2} \quad (17)$$

The calculation method for laser power density E is as follows:

$$E = \frac{P}{\pi \omega(F)^2} = \frac{4f_c^2 P}{\pi \Phi^2 f^2 \cdot \left[1 + \left(\frac{4F\lambda M^2 f_c^2}{\pi f^2 \Phi^2} \right)^2 \right]} \quad (18)$$

The values of the relevant constants are shown in the table below.

The laser energy densities corresponding to the several defocus values mentioned in this article are shown in the table below.

The empirical constants presented in the functional expression for energy density (Fig. 2(g)) are analytically determined via linear regression of the adaptive welding parameters outlined in Table 5. Specifically, the real-time laser power (P) and defocusing distance (F) exhibit strict linear correlations with the gap width (G): $P = 1100 + 1000G$ (W) and $F = 4 + 5G$ (mm). Incorporating the inherent optical system parameters (Table 4), the beam waist radius (ω_0) and Rayleigh length (Z_R) are calculated to be $39.4\mu\text{m}$ and 2.36mm , respectively. By substituting these linear correlations into the fundamental energy density equation, $E = P / \left(\pi \cdot [\omega(F)]^2 \right)$, the integrated mathematical expression depicted in Fig. 2(g) is rigorously formulated.

Table 4. Laser welding constant data.

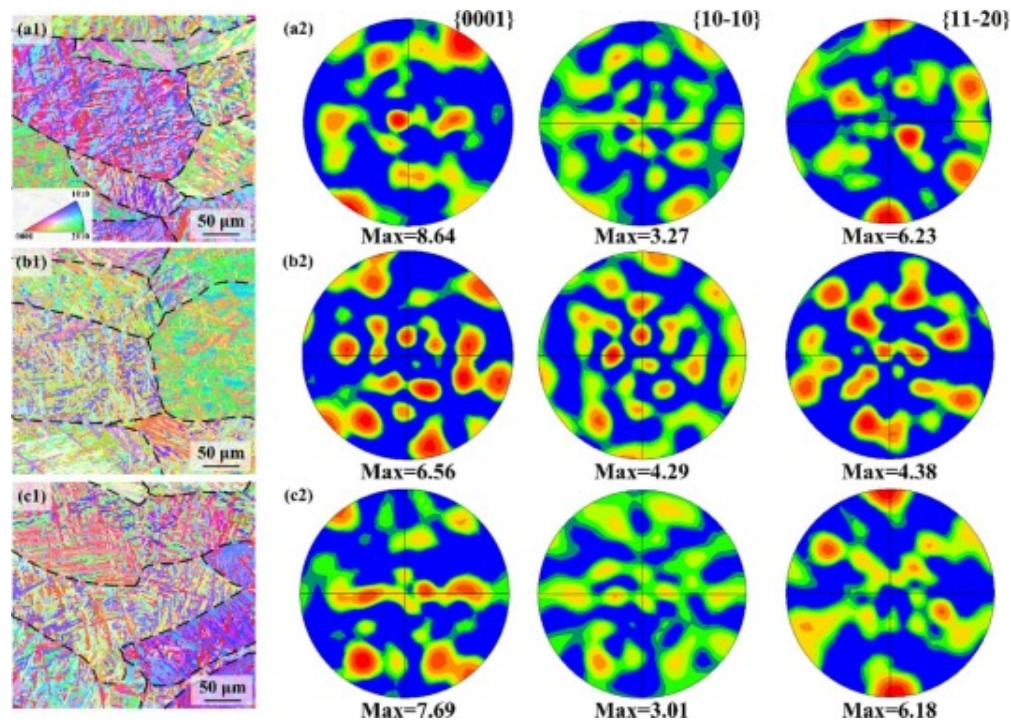
Focal length of the focusing mirror f/mm	Collimator focal length f_c/mm	Fiber core diameter Φ/mm	Laser wavelength λ/nm	Beam quality factor M^2
150	150	100	1030	2

Table 5. Laser welding parameters for different positions.

Laser power/W	Defocus amount/mm	Gap/mm	Spot radius/mm	Laser energy density $E/\text{kW}\cdot\text{mm}^{-2}$
1100	4	0	0.0774	58.446
1200	4.5	0.1	0.0848	53.117
1300	5	0.2	0.0922	48.677

Governed by the dynamic modulation of spatial energy input, the microstructural evolution across the weld initiation region exhibits distinct morphological characteristics, as illustrated in Fig. 2(b, d, and f). The elevated energy density inherent to the initial welding stage triggers the formation of a characteristic X-shaped fusion zone. Within this highly localized domain, primary β columnar grains nucleate epitaxially from the partially melted equiaxed grains at the fusion boundary, propagating inward parallel to the maximum thermal gradient. As further elucidated in Fig. 3(b), these β columnar grains undergo competitive growth along two divergent trajectories bifurcated by the midpoint of the fusion line: the upper grains converge steeply toward the top-center of the fusion zone, whereas the lower grains progress toward the bottom-center. Notably, the columnar structures in the lower region are morphologically finer and display a significantly shallower growth orientation compared to the steeper, coarser grains occupying the upper domain. Furthermore, the large-scale prior- β grain boundaries, delineated by black dashed lines in Fig. 3(a1–c1), define the high-temperature

microstructural framework established immediately post-solidification. The apparent size discrepancy between these coarse prior- β envelopes and the finer internal features (Fig. 2) arises because these parent β grains undergo a rapid, diffusionless phase transformation during cooling, resulting in the fine, intra-granular α' martensitic structures observed at room temperature.



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Fig. 3. EBSD maps depicting grain orientation and pole figures: (a1) Grain orientation at Position #1; (a2) Pole figure at Position #1; (b1) Grain orientation at Position #2; (b2) Pole figure at Position #2; (c1) Grain orientation at Position #3; (c2) Pole figure at Position #3.

As the welding process progresses, the assembly gap increases, leading to an enlarged laser spot radius and reduced power density. As shown in Fig. 2(d), the weld seam widens

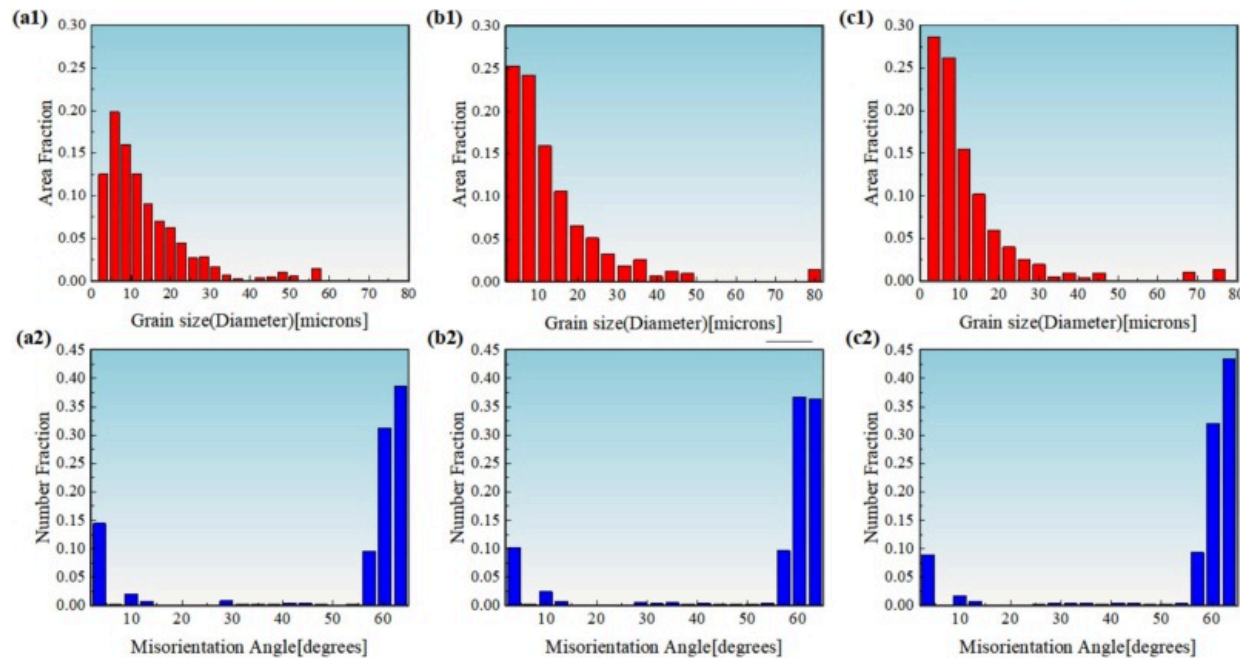
on the front side and narrows on the back side, producing a concave profile in the central fusion line. Although the overall distribution of β columnar grains within the fusion zone remains unchanged, their size increases slightly due to competitive growth. According to solidification theory, columnar grain growth is strongly influenced by the degree of undercooling. During laser welding of Ti-6Al-4V, the high cooling rate induces significant undercooling, which promotes rapid growth of β columnar grains with a strict crystallographic orientation. Because the β phase of titanium alloy has a body-centered cubic structure, it preferentially grows along the $\langle 100 \rangle$ direction.

Because the orientations of partially melted grains at the fusion line differ, β columnar grains in the fusion zone exhibit preferential growth. Grains whose $\langle 100 \rangle$ axis is perpendicular to the molten pool isotherm-aligned opposite to the maximum heat dissipation direction-grow rapidly, while those with other orientations are suppressed. When neighboring grains share the same orientation, they coalesce to form larger grains that continue to grow. At the final stage of welding, where the assembly gap reaches its maximum and the laser power density is lowest, the weld seam develops a Y-shaped profile. The lower section of the fusion line becomes nearly perpendicular to the weld's bottom surface, and except for β columnar grains at the bottom of the fusion zone, all other grains grow toward the center of the upper surface.

EBSA analysis of the three samples (Fig. 3) shows that α' martensite develops along $\langle 0001 \rangle$, $\langle 10\bar{1}0 \rangle$, and $\langle 11\bar{2}0 \rangle$ directions, with texture intensity strongest along $\langle 0001 \rangle$. This indicates a preferential growth orientation under varying gap conditions. At Position #2, texture intensity is weaker and orientations more randomized, reflecting competitive growth along the direction of maximum cooling rate. Positions #1 and #3 maintain stable weld seam morphology, whereas Position #2 forms a transition zone where the seam shifts from X-shape to Y-shape. Abrupt thermal flow variations in the molten pool promote increased randomness in crystal growth.

Fig. 4 illustrates the grain size distribution and grain boundary misorientation in the fusion zone of Ti-6Al-4V alloy under different gap conditions during adaptive welding.

The average martensitic grain sizes are 13.5 μm , 13.4 μm , and 12.2 μm at the respective locations, showing a gradual reduction with welding progression. This reduction is primarily attributed to martensitic phase transformation, in which grain size is mainly controlled by the laser energy density (E).



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Fig. 4. Distribution of grain size and grain boundary misorientation for adaptive weld joints under varying gap conditions: (a1) α' martensite grain size in the fusion zone at position #1; (a2) Grain boundary misorientation in the fusion zone at position #1; (b1) α' martensite grain size in the fusion zone at position #2; (b2) Grain boundary misorientation in the fusion zone at position #2; (c1) α' martensite grain size in the fusion zone at position #3; (c2) Grain boundary misorientation in the fusion zone at position #3.

Calculations indicate that increasing the gap, laser power, and defocusing amount progressively reduces the laser energy density (E), thereby decreasing martensitic grain size within the fusion zone. The $\beta \rightarrow \alpha/\alpha'$ phase transformation in Ti-6Al-4V alloy follows the Burgers orientation relationship, which enables the uniform formation of 12 variants under indistinguishable transformation conditions. The misorientation angles of these variants are mainly concentrated at 10.53° , 60° , 60.83° , 63.26° , and 90° [26]. (See Table 6). Under varying gap conditions, the measured misorientation angles are predominantly distributed around 3.575° , 57.125° , 60.275° , and 63.425° . To quantify the degree of variant selection (DVS) of the α phase during solidification, Qiu et al. [27] proposed the following equation:

$$DVS = \sum_{i=1}^6 |P_L(i) - P_S(i)|$$

here $P_L(i)$ and $P_S(i)$ represent the proportions of a specific variant aligned in the same direction relative to the total variant population for the theoretical and experimental cases, respectively. The table below summarizes the theoretical orientations of α -phase variants in titanium alloys and their corresponding proportions.

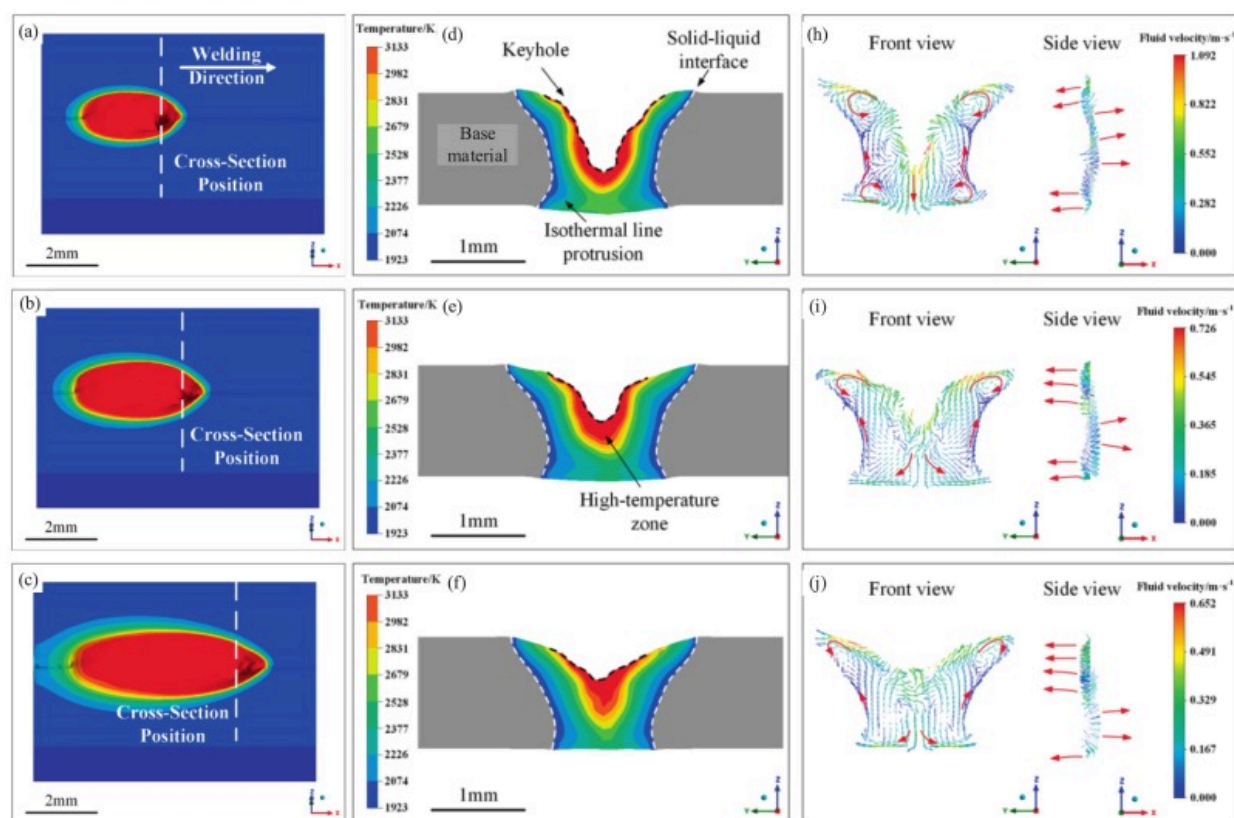
Table 6. Types of misorientations between different α variants and the corresponding probabilities without variant selection.

Misorientation types	Rotation axis/angle	$P_{\text{random}}(\%)$
1	I(identity)	8.3
2	$[1 \ 1 \ \bar{2} \ 0]/60^\circ$	16.6
3	$[1 \ 0 \ \bar{7} \ 17 \ 3]/60.83^\circ$	33.3
4	$[10 \ 5 \ 5 \ 3]/63.26^\circ$	16.6
5	$[7 \ 17 \ 10 \ 0]/90^\circ$	16.6
6	$[0 \ 0 \ 0 \ 1]/10.53^\circ$	8.3

The results indicate that the Degree of Variant Selection (DVS) in the fusion zones of Ti-6Al-4V adaptive laser welds at different positions are 0.6016, 0.5455, and 0.5902, respectively. The decrease in variant selection ratio reflects a more random solidification process, primarily due to modifications in thermal flow behavior induced by changes in molten pool morphology, which facilitate stochastic growth of the α -phase grains.

4.2. Thermal-flow distribution characteristics

The transient evolution of the keyhole plays a deterministic role in modulating the fluidic trajectories within the molten pool. To clarify the correlation between keyhole instability and subsequent microstructural characteristics, the coupled temperature–velocity distributions in the keyhole-adjacent regions were scrutinized. [Fig. 5](#) illustrates the temporal evolution of the molten pool surface morphology during the adaptive laser welding of Ti-6Al-4V joints featuring linear gap variations, alongside the designated sampling planes for transverse analysis. The corresponding thermal profiles and flow field characteristics across these representative cross-sections are detailed in [Fig. 5\(a–c\)](#).



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Fig. 5. Temperature and flow field distributions within the molten pool cross-section at different time intervals in the keyhole region: (a–c) Position #1, #2, #3. (d–f) Temperature field distributions; (h–i) Flow field distributions.

The temperature field of the molten pool was first analyzed. At the early welding stage, the high laser power density formed a deep keyhole, producing a steep temperature gradient, particularly along the thickness direction. The 2377K isotherms crossing the upper and lower molten pool surfaces exhibited a pronounced convex shape. As the heat source advanced toward the central region, the keyhole depth decreased, causing the lower solid–liquid interface to contract inward and modifying the temperature

distribution. The temperature gradients in both the thickness direction and toward the adjacent base metal declined, while the isothermal line penetrating the molten pool decreased from 2377K to 2226K. In addition, a large high-temperature zone appeared beneath the keyhole tip compared with the earlier stage.

At the final stage of unequal-gap adaptive laser welding, the keyhole retained only minimal depth necessary to sustain molten pool penetration, while the lower solid–liquid interface continued to contract inward. As a result, the molten pool shape evolved from an “X” to a “Y” profile. The temperature gradient within the molten pool further decreased, with the largest reduction along the thickness direction. Although the isothermal line penetrating the molten pool remained at 2226K, its width narrowed substantially compared with earlier stages.

The temperature and its gradient within the molten pool are mainly governed by the laser energy density and absorption rate. Higher energy density and absorption increase the molten pool temperature and steepen the gradient. Nevertheless, the temperature field distribution is not determined solely by these factors. The temperature distribution controls the direction and velocity of fluid flow, while the dynamic fluid flow in turn modifies the temperature field [34].

Fig. 5(h) shows the flow field distribution from two views of the molten pool under the deep keyhole effect. High-speed Marangoni circulations are evident near both the upper and lower surfaces, with velocities ranging from 0.552 to $0.822\text{ m}\cdot\text{s}^{-1}$. Their driving forces, however, differ: surface tension dominates on the upper surface, while recoil pressure drives the lower circulation. The fluid beneath the keyhole tip, accelerated by recoil pressure, moves at $1.092\text{ m}\cdot\text{s}^{-1}$ toward the center of the lower surface. On both sides, fluid flows obliquely downward with comparable high-speed characteristics, sustaining the lower Marangoni circulation. A small fraction of fluid near the surfaces detaches from the circulation and drifts slowly toward the molten pool center, eventually bypassing the keyhole wall and advancing to the leading edge. Overall, the cross section exhibits a

pattern in which the upper and lower flows move rearward, whereas the central flow advances toward the leading edge.

At the moment shown in Fig. 5(i), the circulation velocity on the upper molten pool surface decreased, with an average of $\sim 0.455 \text{ m} \cdot \text{s}^{-1}$ and a maximum of $0.726 \text{ m} \cdot \text{s}^{-1}$. In contrast, the lower-surface Marangoni circulation nearly disappeared. Beneath the keyhole tip, a high-velocity zone persisted, but its flow direction became irregular, producing turbulence. The convergence of multiple high-speed streams generated a local high-temperature state and caused variations in the cooling gradient, promoting random crystal growth. Part of the fluid flowed downward and diverged laterally, while most lost its high-speed character and could not form circulation under surface tension alone. Consequently, the majority of fluid moved rearward, with a smaller fraction ascending along the solid–liquid interface toward the upper molten pool.

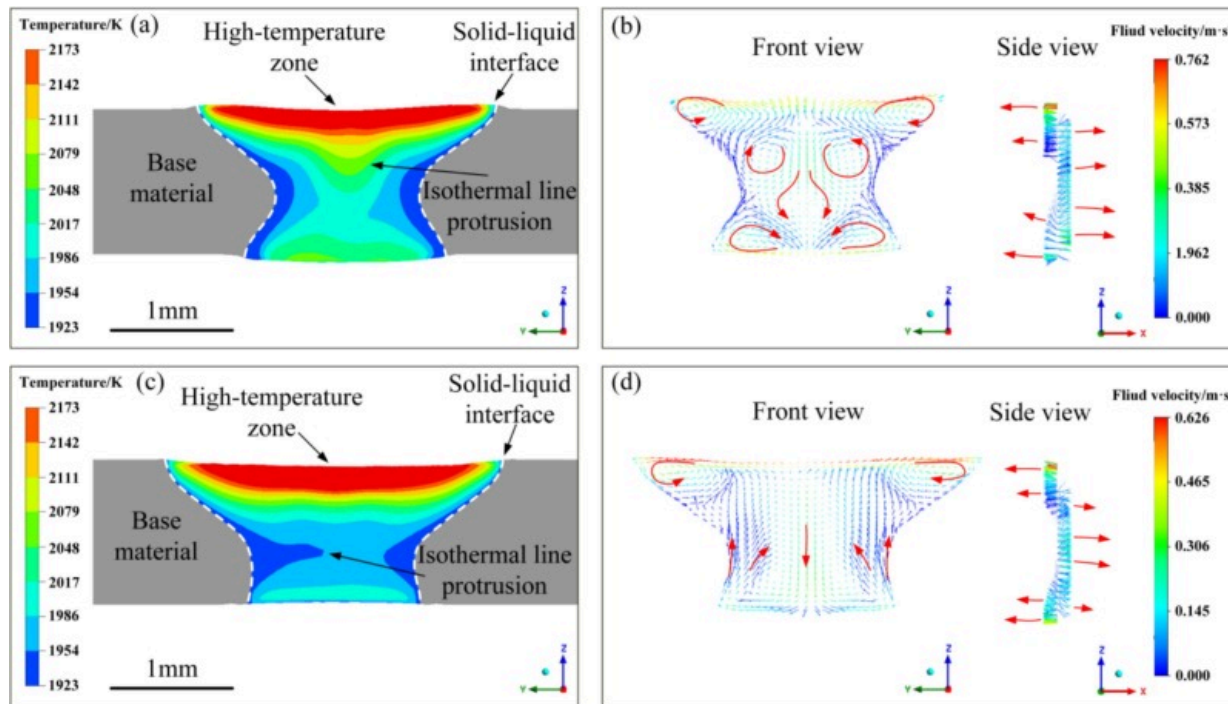
As shown in Fig. 5(j), when the laser power density decreases further, the overall fluid flow velocity in the molten pool is further reduced. A region of high-speed flow reappears on the upper surface, with a maximum velocity of $0.652 \text{ m} \cdot \text{s}^{-1}$. At this stage, circulation is confined to the upper surface, while turbulence persists at the tip of the keyhole, although its intensity and disorder are less pronounced than before. The downward flow toward the bottom of the molten pool exhibits a lower average velocity of $0.329 \text{ m} \cdot \text{s}^{-1}$. The flow pattern shows that nearly the entire upper half of the molten pool moves toward the rear, while most of the fluid in the lower half flows toward the front, except for the region near the lower surface.

Therefore, the variation in the uniformity of the molten pool temperature field distribution in the keyhole region during Ti-6Al-4V alloy laser welding with unequal-gap can be reasonably explained. At the initial stage of welding, the high laser power density provides a strong driving force for molten pool convection, leading to high-speed, high-energy, and well-organized fluid motion. This promotes rapid and uniform heat redistribution within the molten pool, thereby increasing the temperature gradient and producing a more uniform and orderly temperature field distribution. Under such strong

driving conditions, high-speed Marangoni circulations form near both the upper and lower surfaces of the molten pool. In this region, instead of dissipating or transferring outward, heat is concentrated within the molten pool, resulting in the bulging of isothermal contours.

In the later stages of welding, as the laser power density decreases, the driving force for molten pool convection weakens, resulting in a marked reduction in fluid velocity. Consequently, Marangoni circulation is no longer sustained on the lower surface of the molten pool. Nevertheless, the bulging of isotherms persists, primarily due to the lateral divergence of fluid in the lower molten pool region. In addition, the convergence and collision of fluid streams in the central molten pool give rise to increasingly complex and disordered flow behavior, which in turn produces a localized high-temperature zone at the center of the molten pool.

For Ti-6Al-4V alloy, both the nucleation and growth of β columnar crystals and the subsequent β -to- α phase transformation are strongly influenced by the temperature gradient and overall temperature field distribution within the molten pool during cooling. To analyze the characteristics of the molten pool temperature field in the cooling stage under varying gap conditions, two representative positions, as indicated in Fig. 8(a) and (c), were selected. The corresponding temperature and flow fields were recorded 60ms after the termination of direct laser heating, and the results are presented in Fig. 6.



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Fig. 6. Comparative analysis of temperature and flow field distributions within the molten pool during the cooling stage of unequal-gap adaptive laser welding: (a) temperature field distribution in the initial segment of the weld seam; (b) temperature field distribution in the terminal segment of the weld seam; (c) flow field distribution in the initial segment of the weld seam; (d) flow field distribution in the terminal segment of the weld seam.

A comparison of the temperature fields was first conducted. Using the waist of the molten pool as the boundary, the upper portion of the pool lies in the high-temperature region, while the lower portion shows a temperature decrease, with the lowest temperature consistently located at the center. After the same cooling duration, the lower half of the molten pool in the final section of the weld cools to 1986K, whereas the corresponding

region in the initial section remains above 2048K. Moreover, the final section exhibits a steeper temperature gradient, indicating a faster cooling rate. This difference in cooling rate is a critical factor contributing to the uneven microstructure morphology and distribution characteristics between the initial and final weld sections during solidification.

Further comparative analysis of the flow field results reveals the following: by examining both the frontal and lateral views, it can be observed that the fluid flow directions within the molten pool cross-sections at the two selected positions can generally be summarized as follows: the fluid flows from the four corners towards the rear of the molten pool, while the fluid in the center flows towards the front. This flow behavior provides a reasonable explanation for the existence of the high-temperature region within the molten pool.

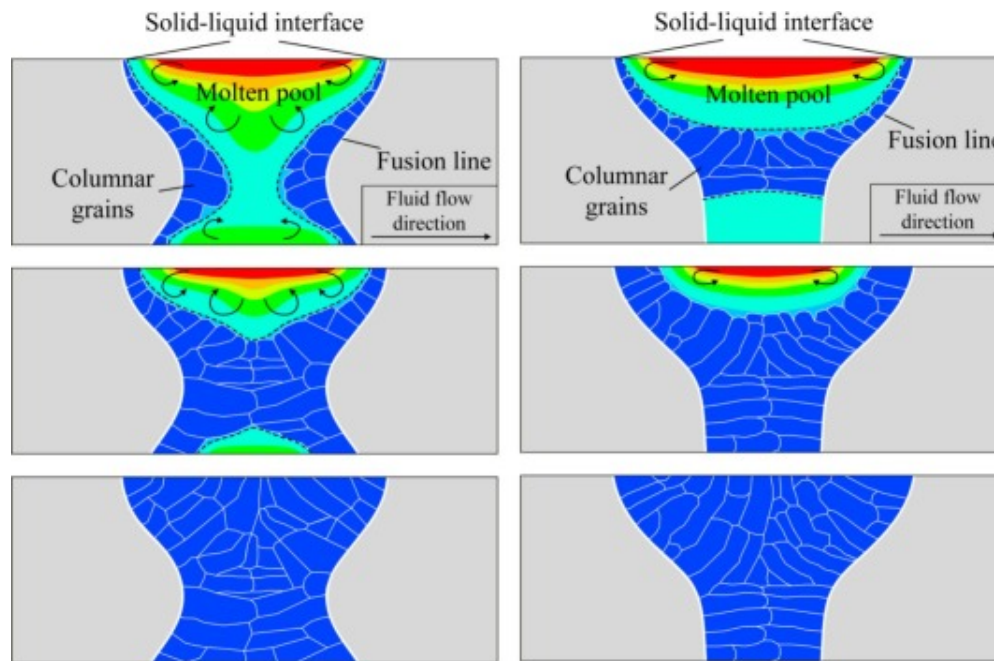
Under the influence of the laser heat source, high-temperature, high-energy fluid forms at the front of the molten pool. This fluid flows toward the rear along the upper surface, generating a high-speed, high-temperature Marangoni circulation that leads to the formation of a localized high-temperature zone near the upper surface. The cooling rate difference between the lower halves of the molten pool in the initial and final weld sections is partly attributed to the circulation beneath the molten pool in the initial section, which inhibits rapid heat dissipation by trapping heat within the fluid flow.

Moreover, the fluid flow behavior at the center of the molten pool differs between the initial and final weld sections. In the initial section, two circulating flows develop in the upper-central region, contributing to the bulging of temperature contours in the high-temperature zone above the pool. In contrast, the central region of the molten pool in the final section contains low-temperature, low-velocity fluid converging toward the center. This flow gradually obstructs the interaction between the upper and lower molten pool fluids during cooling, leading to the bulging of temperature contours in the low-temperature zone, as shown in Fig. 6(b). Ultimately, these low-temperature regions coalesce at the molten pool center.

4.3. Influence mechanism of molten pool heat-flow behavior on microstructural uniformity

Building on the previous analysis of microstructural morphology, distribution characteristics, and the associated temperature and flow field distributions in Ti-6Al-4V alloy under unequal-gap adaptive laser welding, this section investigates the influence of molten pool flow behavior on microstructural uniformity.

Initially, the study examines the dendritic growth patterns during the unequal-gap adaptive laser welding of Ti-6Al-4V alloy, as shown in [Fig. 7](#). The dendritic growth displays a clear correspondence with the weld profile. In essence, dynamic variations in molten pool flow behavior during welding give rise to differences in the morphology and uniformity of the dendritic structure. At the onset of solidification and cooling in the initial weld region, multiple circulations in the upper and lower parts of the molten pool elevate the local temperature, allowing the pool to remain continuous. Dendrites nucleate and grow vertically from the pool center toward the central axis, initiating along the midline.



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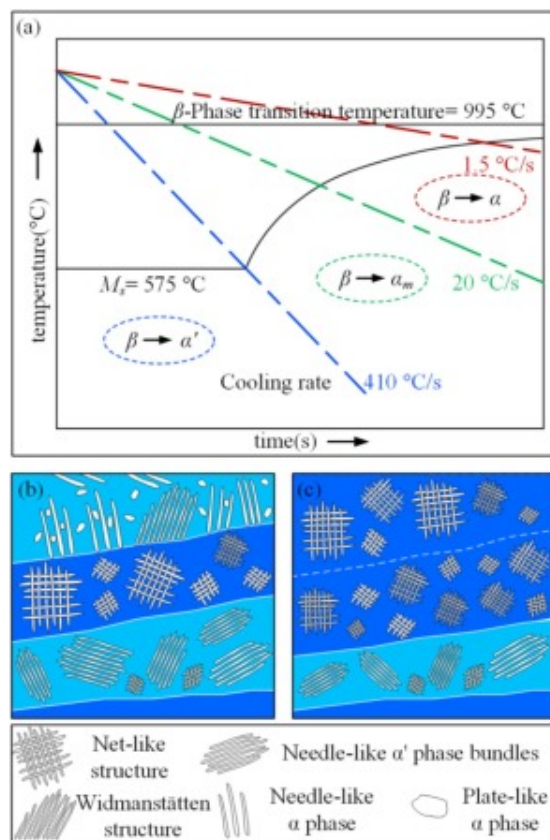
Fig. 7. Schematic illustration of columnar crystal growth during the solidification stages in the initial and terminal segments of the weld seam in Ti-6Al-4V alloy unequal-gap adaptive laser welding: (a–b) early stage of solidification; (c–d) intermediate stage of solidification; (e–f) final stage of solidification.

During the mid-cooling stage, dendrites in the upper part of the molten pool initially grow toward the central axis but reorient toward the surface as they approach the center, forming growth corners. This phenomenon arises from molten pool circulation, which modifies the temperature field distribution, causing isotherm bulging, altering the temperature gradient direction, and thereby redirecting dendritic growth. In contrast, dendrites in the lower part of the pool grow at a fixed angle toward the bottom surface. Circulation in this region does not significantly alter the temperature field but primarily influences the cooling rate. The isotherms in the lower pool show no obvious bulging,

preventing the formation of growth corners in downward-growing dendrites. Moreover, the growth angle of these dendrites is noticeably more gradual than that of dendrites in the upper region, as shown in [Fig. 7\(c\)](#).

In the final segment of the weld, the dynamic variations in molten pool flow behavior alter both the pool shape and the characteristics of dendritic growth. Unlike the X-shaped molten pool observed at the weld's start, the Y-shaped molten pool at the end exhibits weaker bottom circulation and slower surface flow. This reduction in fluid motion accelerates heat dissipation, producing lower temperatures and higher cooling rates. During the early stage of solidification, dendrites still initiate growth from the pool center, effectively separating the upper and lower molten pool regions. However, in the mid-solidification stage, the absence of central circulation leads to a flattened temperature field distribution. Without bulging isotherms at the pool center, dendrites primarily grow along the temperature gradient, resulting in a larger growth angle but without forming pronounced growth corners. By the completion of solidification, the weld's final segment develops smaller dendritic structures, a consequence of the steeper temperature gradient and faster cooling rate.

Upon completion of the thermal cycle, the solidified molten pool initially retains a temperature profile exceeding the β -transus temperature ($T_{\beta} \approx 995^{\circ}\text{C}$). At this elevated thermal state, the dendritic architecture is predominantly constituted by the high-temperature β -phase. As the temperature dissipates, the ultimate microstructural morphology is dictated by the local cooling rate (T), following the kinetic pathways delineated in the Continuous Cooling Transformation (CCT) diagram ([Fig. 8](#)).



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Fig. 8. Schematic illustration of the CCT curve and corresponding phase evolution/distribution in the fusion zone during unequal-gap adaptive laser welding: (a) Continuous cooling transformation (CCT) curve of Ti-6Al-4V alloy; (b) Phase composition and distribution in the initial segment of the weld; (c) Phase composition and distribution in the terminal segment of the weld.

The competitive nature of these phase transitions is summarized as follows:

Rapid Quenching ($T > 410^{\circ}\text{C/s}$): The prior β -grains undergo a diffusionless transformation, resulting in a fully acicular martensitic α' phase.

Moderate Cooling ($20 \leq T \leq 410^\circ\text{C/s}$): A diffusion-controlled nucleation and growth process prevails, leading to the formation of a characteristic lamellar α morphology.

Near-Equilibrium Cooling ($T \leq 20^\circ\text{C/s}$): The system permits sufficient diffusive transport for the β -phase to transition into the thermodynamic equilibrium $\alpha + \beta$ microstructure.

In the weld seam of Ti-6Al-4V during unequal-gap adaptive laser welding, columnar grains often exhibit alternating light and dark regions, with most grain boundaries appearing weakened or indistinct. This feature is closely related to β -phase transformation during cooling. In the initial weld segment, high laser power density produces greater heat input, together with dynamic molten pool flow, induces spatial variations in cooling across the fusion zone. The high-temperature β -phase rapidly transforms into fine acicular α' , which intersect within grains to form a basketweave microstructure. This intersecting arrangement induces phase segmentation and isolation, manifesting as dark relief under microscopic observation. In some regions, however, incomplete β -phase transformation caused by rapid cooling leaves residual β at α' boundaries, enhancing the darker contrast of these areas.

Acicular α' phases precipitating in parallel bundles within grains prevent large-scale phase boundary isolation, producing a microstructure with few dark lines and predominantly bright regions. In some areas, the high-temperature β -phase transforms into lamellar or acicular α instead of acicular α' . The lamellar α disperses within grains, whereas the acicular α nucleates at grain boundaries and grows inward. When the acicular α phases align and cluster, they form coarse Widmanstätten structures, in which densely arranged phase boundaries appear as dark regions and the remaining areas are occupied by white lamellae. Thus, the microstructure consists of dark grain-boundary regions interspersed with alternating white lamellae and dark interstitial areas.

In the terminal weld segment, reduced laser power density accelerates cooling in the fusion zone. Consequently, most of the high-temperature β -phase transforms into fine acicular α' phases, forming smaller basketweave structures or clustered bundles. The

precipitation of lamellar and acicular α decreases markedly, yielding a more uniform and refined microstructure in the fusion zone.

5. Conclusions

This study investigates the influence of varying gap sizes on microstructural evolution and fluid flow behavior during laser welding through combined experimental and simulation approaches. The key findings are summarized as follows:

- (1) As the welding gap increases from 0 to 0.2mm, the weld seam shape transitions from X to Y. β columnar grains, governed by competitive growth, preferentially grow along the $\langle 100 \rangle$ direction, with their average width progressively decreasing.
- (2) At the critical point of the X-to-Y transition, the weld texture strength and α -variant selection in titanium alloys decrease markedly due to abrupt changes in molten pool thermal flow. The average martensite grain size decreases by 9.6% as laser energy density is reduced.
- (3) The temperature field and fluid flow within the molten pool vary markedly with gap size. As the gap increases, the isotherms flatten and the overall temperature decreases, leading to reduced fluid velocity and the disappearance of Marangoni convection at the lower surface.
- (4) A larger welding gap produces a steeper temperature gradient during cooling, resulting in a higher cooling rate. Consequently, under different gap conditions, variations in the growth direction of columnar grains and refinement of martensitic grain size are observed.

CRedit authorship contribution statement

Chao Ma: Writing – review & editing, Writing – original draft, Visualization, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.
Shuhao Tian: Validation, Resources, Investigation. **Shijie Luo:** Data curation, Software, Writing – review & editing. **Yanqiu Zhao:** Supervision, Resources, Methodology, Data curation. **Yue Li:** Visualization, Validation, Investigation. **Xiaohong Zhan:** Visualization, Validation, Resources, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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